Class Diagram – Food Supply Chain Traceability & Provenance Portal (W3-TV2)

Organization
- organizationId: Integer
- name: String

User
- userId: Integer
- name: String
- email: String
- passwordHash: String
- role: Role
- organizationId: Integer
- status: AccountStatus

+ login(): Token
+ hasPermission(action: String): Boolean

AuditLog
- auditId: Integer
- actorId: Integer
- action: String
- entityType: String
- entityId: Integer
- oldValue: JSON
- newValue: JSON
- createdAt: DateTime

+ append(): void {append-only}

Organization 1 — 0..* User
User 1 — 0..* AuditLog
User 1 — 0..* Batch : creates
User 1 — 0..* SupplyChainEvent : records
User ..> «enumeration» Role : uses
User ..> «enumeration» AccountStatus : uses

Batch
- batchId: Integer
- batchCode: String {unique}
- productName: String
- origin: String
- quantity: Decimal
- creationDate: DateTime
- currentStatus: EventType
- createdBy: Integer

+ createBatch(): Batch
+ updateStatus(event: EventType): void
+ validateTransition(next: EventType): Boolean

Batch 1 — 1 QRCode

QRCode
- qrId: Integer
- batchId: Integer
- qrData: String
- signatureHMAC: String
- createdAt: DateTime

+ generate(batchId: Integer): QRCode
+ verifySignature(): Boolean

Batch 1 — 1..* SupplyChainEvent

SupplyChainEvent
- eventId: Integer
- batchId: Integer
- actorId: Integer
- eventType: EventType
- timestamp: DateTime
- location: String
- description: String

+ recordEvent(): void
+ isSequenceValid(): Boolean

SupplyChainEvent ..> «enumeration» EventType : uses

«enumeration» **EventType**
- CREATED
- TRANSPORTED
- RECEIVED
- PROCESSED
- PACKAGED
- SHIPPED

«enumeration» **Role**
- FARMER
- DISTRIBUTOR
- PROCESSOR
- CONSUMER
- ADMIN

«enumeration» **AccountStatus**
- PENDING
- ACTIVE
- LOCKED